Turn-Aware Streaming ASR in One Small Open Model:
Causal Labels for End-of-Turn Detection

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Abstract

Every voice interface has to answer one deceptively hard question in real time: *has the user finished talking?* Commercial systems are converging on *turn-aware streaming ASR* — a single model that transcribes while deciding end-of-turn from semantics rather than from a silence timeout (Deepgram Flux, OpenAI's semantic VAD) — but every member of the class is closed, and nothing about how to train one has been published. We build one in the open: Qwen3-ASR-0.6B, LoRA-fine-tuned to emit an end-of-turn marker inside its transcription stream, trained on one GPU from $\sim$12k synthesized examples. Getting there took eight failed versions spent chasing an apparent recall-vs-precision trade-off: checkpoints good at firing were bad at holding, checkpoints good at holding would not fire, and training runs oscillated between the two modes. The frontier never existed. It was manufactured by *clairvoyant labels* — supervision whose correct value depends on audio *after* the decision point — injected in two distinct ways by offline evaluations and the training pools derived from them. Rebuilding the data under a single principle, **every label must be computable from the causal prefix**, with architecture, recipe, and scale unchanged, eliminates the oscillation and yields one checkpoint that dominates all eight predecessors on a deployment-matched streaming benchmark: 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute, no inference-time crutches. Fixed silence timeouts — the deployed alternative — need 3$\times$ the latency and 5$\times$ the false fires to match that recall, and a fresh 2$\times$-larger test set confirms the operating point. The same model delivers a capability its base model reports no numbers for — a text context prefix lifts hotword recall by +28.9 pp on Earnings-22, an advantage that stays flat as the session grows — at a measured, attributed cost: +1.1/+2.7 pp offline WER from the fine-tune itself (not marker interference; we verify by ablation), with streaming WER unaffected. Everything runs on stock vLLM at 84 ms per 0.5 s chunk. The failure mode we diagnose is not specific to speech: any streaming decision task trained on offline-labeled data can inherit it. Stop training streaming models to see the future.

Code: https://github.com/19PINE-AI/turn-aware-asr | Website: https://01.me/research/turn-aware-asr

1 Introduction

A voice agent that cannot tell when you have finished speaking is unusable in two symmetric ways: it interrupts you mid-thought, or it stares at you after you stop. As full-duplex assistants proliferate, this endpointing decision — fire an end-of-turn event now, or wait — has become

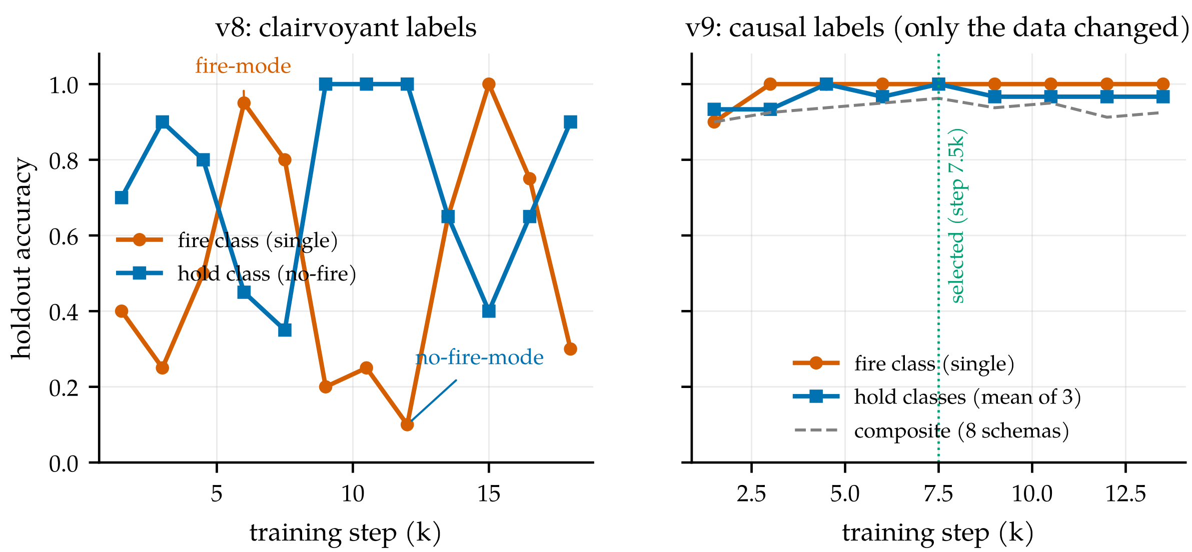

Figure 1. The fingerprint of contradictory supervision. Left: v8's holdout accuracy oscillates for the entire run between a *fire-mode* attractor (fire class ≈1.0, hold class ≈0.4) and a *no-fire-mode* attractor (fire ≈0.1, hold ≈1.0): the training pools assign opposite targets to causally identical audio, so no single policy satisfies both, and optimization circulates between the modes. Right: v9 — same architecture, same LoRA recipe, same data volume, *only the labels changed* — climbs monotonically, and both classes sit at ceiling *simultaneously* from step 3k on.

the load-bearing interaction primitive. The industry default remains a cascade (a voice-activity detector plus a fixed silence timeout), but frontier commercial systems have moved the decision *into* the recognizer: Deepgram's Flux emits `EagerEndOfTurn` events from the ASR model itself [Deepgram, 2026], and OpenAI's Realtime API offers a `semantic_vad` mode [OpenAI, 2025]. We call this class **turn-aware streaming ASR**: one model that transcribes while deciding, from semantics and silence jointly, whether the turn is over. Every member of the class is closed — no weights, no recipe, no account of what makes the training work.

This paper builds a turn-aware streaming ASR model in the open, and reports the one thing about training it that we believe the closed builders each rediscovered privately: a failure mode in how streaming decisions get *supervised*, told through the eight model versions it cost us. We fine-tune a small open unified ASR model (Qwen3-ASR-0.6B, Shi et al., 2026) to emit an endpoint marker token inside its transcription stream, so one model transcribes, detects end-of-turn, and accepts a biasing context prefix. Versions v1–v8 traced out what looked like a fundamental trade-off: checkpoints that fired promptly at turn ends also fired spuriously mid-utterance; checkpoints that never fired spuriously would not fire at all. Holdout curves *oscillated* between the two modes during training (Figure 1, left). The project record at v8 concluded the behaviors lay on a Pareto frontier and recommended shipping two checkpoints, one per deployment.

The frontier was a mirage. Both the offline evaluations and the training pools derived from them contained **clairvoyant labels**: targets whose correct value is a function of audio that arrives *after* the moment the streaming model must act. Two independent instances (§3, Figure 3): (i) offline evaluation clips are cut at the end of speech, so "complete utterance, no trailing silence" is labeled *fire* by the offline pool and *hold* by the streaming pool — opposite targets on causally identical prefixes; (ii) "pause inside a turn" vs. "pause between turns" were labeled by *who speaks next*, information that does not exist at decision time. A model trained on such pools is not underfitting; it is faithfully fitting a contradiction, and it expresses the contradiction as bimodal training oscillation and as phantom trade-offs across checkpoints.

The fix required no architecture change, no scale, and no new data source: rebuild the pools under one rule — *fire iff the speech so far is semantically complete AND* $\geq$0.3 *s of silence has elapsed*, both observable from the prefix — and keep everything else fixed. The oscillation disappears (Figure 1, right). On a deployment-matched streaming benchmark (§4), the single resulting checkpoint dominates all eight predecessors and the VAD-plus-timeout family that products actually deploy (§5, Figure 7).

Our contributions:

1. **A diagnosis with receipts** (§3): two concrete mechanisms by which offline-derived supervision demands clairvoyance from streaming models, and the observable artifacts they produce — training-time mode oscillation and spurious cross-checkpoint trade-offs.
2. **A deployment-matched streaming benchmark** (§4): replayed real conversational timelines with a causally-decidable boundary taxonomy. Under it the historical ranking *inverts* (Figure 5), and a universal blind spot surfaces: every prior model hallucinates on silence-initial audio (Figure 4).
3. **A causal label recipe** (§5) whose minimal-pair construction (Figure 6) turns the former contradiction into the intended discriminative feature, yielding 0.97 recall at 0.39 s P50 and 0.3 false fires/min with no inference-time crutches — confirmed on a fresh 2$\times$-larger test set (0.92 recall, 0.42 s, 0.2 false/min).
4. **Numbers for a capability with no public numbers** (§6): the context prefix lifts hotword recall by +28.9 pp on Earnings-22, flat across session age (Figure 8). The fine-tune's offline WER cost is measured *and attributed* (+1.1/+2.7 pp; a marker-ablation rules out decode truncation).
5. **A serving reality check** (§7): the system runs on stock vLLM at 84 ms per 0.5 s chunk; we document an out-of-distribution landmine in multimodal chunked feeding (88% WER) and its fix (Figure 9).

Section 8 compresses the lesson into a four-question checklist. The pattern — offline-labeled data supervising an online decision — recurs in dialogue turn-taking, barge-in, simultaneous translation, streaming TTS, and agent action-timing. We suspect phantom frontiers are being chased in several of these right now.

2 One model that transcribes, endpoints, and takes hints

Task. A single-channel audio stream arrives in 0.5 s chunks. At each chunk boundary the system must (a) extend a running transcript and (b) decide whether the turn has ended — an irreversible, latency-sensitive event that downstream components act on. A text context prefix (user profile, hotword list) may be supplied at session start and should bias recognition throughout.

Model. We start from Qwen3-ASR-0.6B [Shi et al., 2026], an open unified speech model (AuT audio encoder + dense Qwen3 decoder) with strong offline WER but no endpointing and

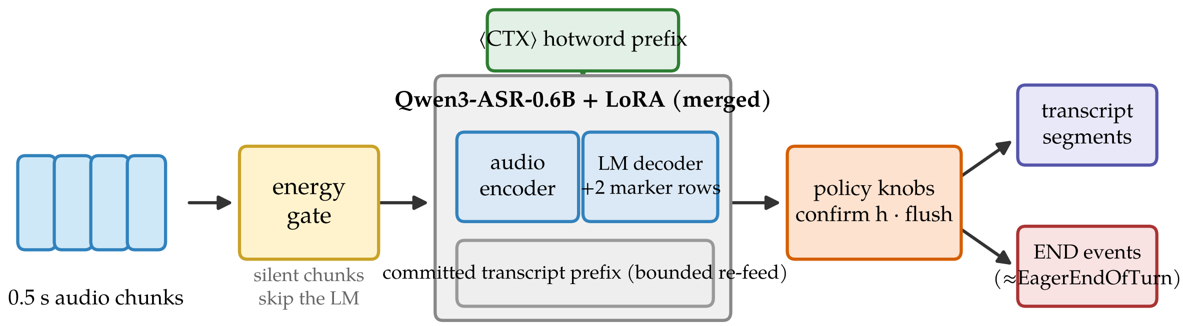

Figure 2. System shape. Audio chunks pass an RMS energy gate (silent chunks never start a segment, so mostly-silent sessions are nearly free); the merged LoRA model transcribes the current bounded segment with the `<CTX>` hotword prefix in the system slot and may emit marker tokens; two policy knobs (confirm horizon, max-segment force-flush) sit *behind* the model. Outputs are transcript segments plus END events — the same event shape as Flux's `EagerEndOfTurn`.

no published context-biasing numbers. We claim two reserved vocabulary slots as marker tokens, `<EAGER_END_SPEECH>` and `<END_SPEECH>`, and LoRA-fine-tune [Hu et al., 2022] the decoder (r=16, α=32, q/k/v/o projections, plus the two marker embedding rows) to emit them inside the transcript at endpoint moments. Training data is ∼12k examples synthesized from LibriSpeech [Panayotov et al., 2015] and AMI [Carletta et al., 2005] (§5); training takes hours on one GPU.

Inference policy. Three orthogonal, interpretable knobs (Figure 2); every reported arm states its knob settings. The **energy gate** never *starts* a segment on a silent chunk — motivated by a data blind spot (§4), and also the serving trick that lets one engine carry many mostly-silent sessions (§7). The **confirm horizon** h holds a fire candidate for h further silent chunks before signaling END (resumed speech cancels the signal; the transcript flush stands) — the phrase-vs-turn dial, which v9 barely needs. The **max-segment force-flush** bounds the committed transcript segment; a transcription reset, never an END event.

Table 1. The landscape. "Cascade" = VAD + silence timeout + separate ASR: the deployed default, and our measured baseline in Figure 7. Flux and `semantic_vad` are the closed members of the turn-aware class this work opens.

	This work	Flux / `sem.vad`	Qwen3-ASR	Kyutai STT	Cascade
Streaming ASR	✓	✓	✓	✓	✓
Semantic end-of-turn	✓	✓	—	—	timeout only
Context biasing (numbers)	✓ (+28.9 pp)	—	unreported	—	ASR-dep.
Open weights + recipe	✓	—	weights only	weights only	✓

3 The phantom frontier

Eight versions were trained before the diagnosis; in hindsight the record reads like a controlled study of what contradictory supervision does to a model. v3, selected by an offline meeting

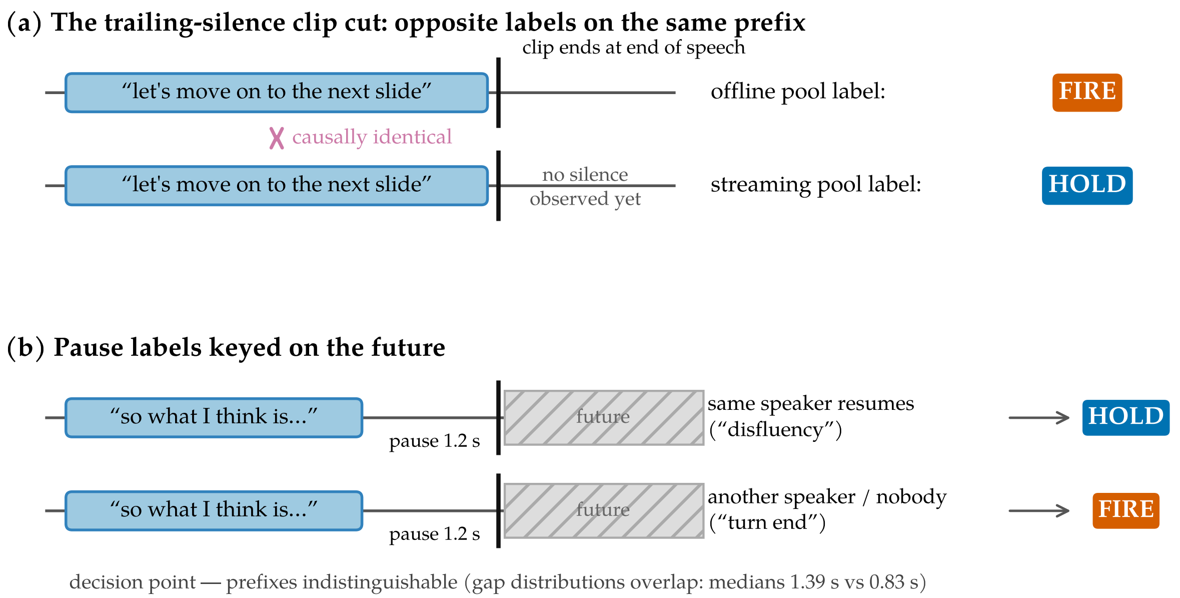

Figure 3. Two ways offline-derived supervision demands clairvoyance. (a) Offline clips inherit forced-alignment cuts at the end of speech, so the offline pool demands a fire on exactly the prefix the streaming pool labels *hold* — and the condition "speech ended, no silence yet" is one deployment never presents, because silence keeps arriving after a real speaker stops. (b) Disfluency-vs-boundary pause labels are keyed on who speaks *next*; at the decision point the prefixes are statistically indistinguishable (the two classes' silence-gap distributions overlap almost completely).

benchmark, fired eagerly and accurately *on that benchmark*. v5 balanced offline and streaming pools and over-fired at phrase boundaries. v7–v8 added a dedicated "do not fire" pool, and training began to oscillate between a fire-mode and a no-fire-mode attractor every few thousand steps, never converging (Figure 1, left). Across checkpoints, recall and precision appeared to trade off so consistently that the v8 postmortem proposed shipping *two* checkpoints.

The oscillation was the tell. A model whose loss cannot be minimized because the data assigns different targets to near-identical inputs does not settle; it circulates among modes that each satisfy one subset of the labels. Two mechanisms injected exactly that (Figure 3):

Contradiction 1: the trailing-silence clip cut. Offline ASR clips end at the end of speech — datasets are segmented by forced alignment, and evaluation clips inherit the cut. An offline endpoint benchmark built on such clips therefore demands a fire on audio that ends during or immediately after speech. The streaming pools (correctly) teach the opposite: never fire before silence is observed. The condition "complete utterance, no trailing silence" thus carries the label *fire* in one pool and *hold* in the other. v8's two pools were disjoint sets of AMI utterances with opposite labels on identically distributed content: 50/50 label noise on the exact decision the model exists to make.

Contradiction 2: labels keyed on who speaks next. The historical pools separated "disfluent pause — hold" from "turn boundary — fire" using the transcript's segmentation: same speaker continues ⇒ disfluency; another speaker follows ⇒ boundary. Measured on the v5 pool, the two classes' silence-gap distributions overlap almost completely (medians 1.39 s vs 0.83 s). At decision time — mid-pause — the classes are indistinguishable; the label is a function of the future. An offline decoder can satisfy such labels by peeking; a causal model cannot, and part of the historical "disfluency over-fire" penalty punished models for lacking clairvoyance.

Clairvoyant labels

A label for a streaming decision at time t is *clairvoyant* if its value depends on input after t. Pools containing clairvoyant labels do not merely add noise: because the dependence is systematic, they partition the data into internally consistent subsets with incompatible optima. The observable symptoms are (a) training-time oscillation between mode attractors and (b) apparent capability trade-offs across checkpoints — a phantom Pareto frontier. **Principle: every training label and every evaluation target for a streaming decision must be computable from the causal prefix.**

4 Measuring what deployment sees

If clairvoyant labels are the disease, evaluation is where it enters. We replaced both legacy benchmarks with one deployment-matched protocol — and applied the causality principle to the benchmark itself.

Material and ground truth. Continuous single-channel stretches from held-out AMI meetings: one target speaker's utterances at their true timeline offsets, silence between them, 30–60 s per stretch (the voice-assistant shape: one microphone, one user; audio never ends at a speech boundary). Each utterance-end boundary is classified by what is observable *on this channel*: **turn-final** (same-speaker gap ≥ 2 s) — should fire; **continuation** (gap in $[0.3, 2)$ s, same speaker resumes) — firing is a measured segmentation *cost* (`resume_after_fire`), not an error, because the correct answer depends on the future; **internal** (pauses inside utterances, or <0.3 s after speech) — firing is a *false fire*. Metrics: boundary recall within $[-0.25, +1.5]$ s, false fires per speech-minute, latency percentiles, resume rate, streaming WER of flushed transcripts, per-chunk compute. An early version of this benchmark itself contained a clairvoyant rule (promoting boundaries to turn-final when *another meeting speaker* interleaved — inaudible on this channel); our own checklist (§8) caught it, and all numbers use the corrected classification.

Finding 1: silence incompetence. Figure 4. Every v1–v8 training example begins with speech; a real channel is mostly silence. On silence the models hallucinate and fire; ungated numbers are uninterpretable, and all comparisons below run gated.

Finding 2: the ranking inverts. Figure 5. A benchmark with clairvoyant targets does not just mis-score models; it mis-*ranks* them, and the error is largest exactly at the top — v3 was "best" offline *because* it fires without waiting for silence, which is the single worst property in deployment.

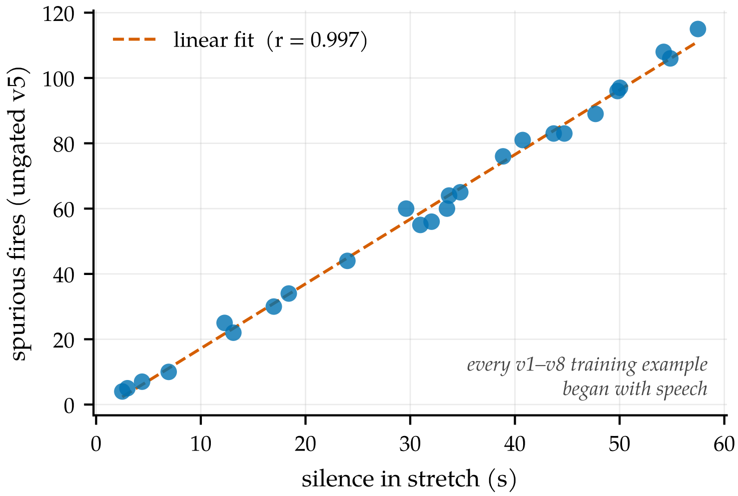

Figure 4. Every prior model is silence-incompetent. Spurious fires of ungated v5 against the silence content of each benchmark stretch: the model hallucinates text and sprays markers on silence at $\sim$1.9 fires/s ($r = 0.997$), because every v1–v8 training example begins with speech while a real channel is mostly silence (13 of 22 minutes here). Consequence: *ungated* recall is uninterpretable — a random 1.9 Hz spammer scores 0.96 recall against the tolerance window. The fix costs one line (RMS energy gate), and v9 additionally bakes silence schemas into training.

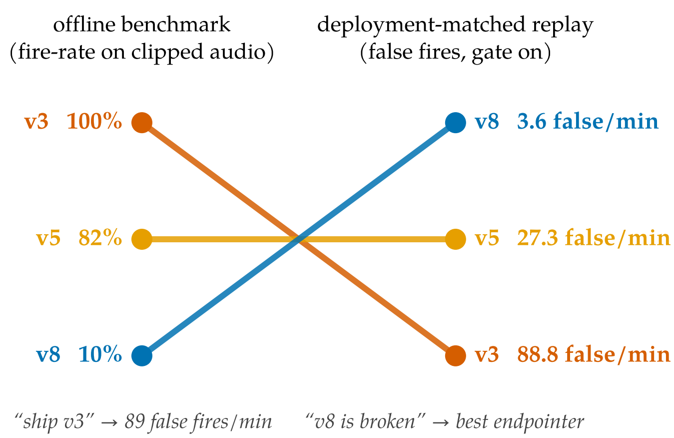

Figure 5. The ranking inversion. Left: the legacy offline benchmark (fire-rate on clips cut at end-of-speech). Right: the deployment-matched replay benchmark (false fires per speech-minute, energy gate on; full metrics in Table 2). The offline champion v3 fires 89 times per speech-minute mid-utterance — its offline strength *is* a premature-fire pathology once audio keeps flowing. v8, written off at 10% offline, is the best prior endpointer. Both historical ship decisions were wrong.

5 Relabel, don't rearchitect

v9 changes only the data. One rule: *emit <EAGER_END_SPEECH><END_SPEECH> at a point iff the speech so far is semantically complete AND* $\geq$*0.3 s of silence has elapsed since the last speech.* Both conditions are functions of the prefix. Two corollaries kill the two contradictions. (1) A complete utterance with no silence tail gets no marker — and the *same utterance* appears again with a silence tail and the marker (Figure 6): where v8 assigned opposite labels to two pools of identically distributed audio (unlearnable), v9 assigns opposite labels to the same audio

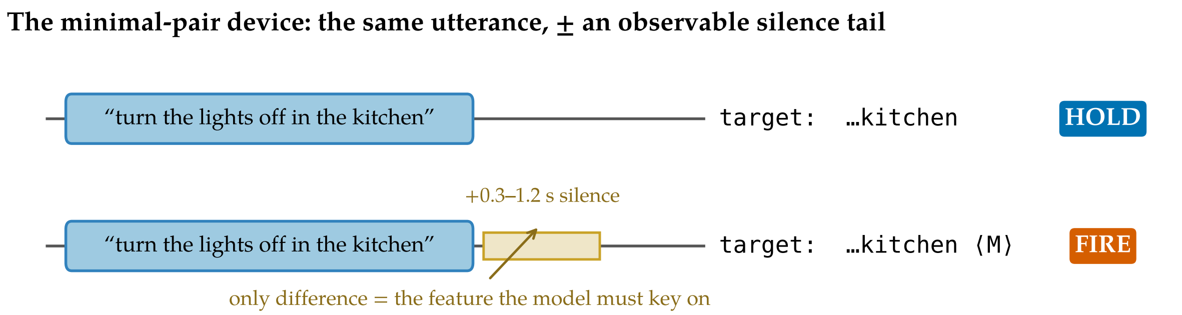

Figure 6. The minimal-pair device. Each complete utterance appears twice in v9's training data: without a silence tail (target: transcript only) and with a 0.3–1.2 s tail (target: transcript + marker). The only difference between the opposite-labeled examples is the observable the model must use. This construction is what removes the oscillation in Figure 1.

Table 2. Main result (deployment-matched replay, energy gate on; latency-vs-false-fire view in Figure 7). Rows 1–4: the 25-stretch development benchmark (96 turn-final boundaries). v9 dominates every predecessor without either crutch: its *raw* false-fire rate beats v5's *confirmed* rate, and its WER needs no force-flush. The confirm ablation: v5's horizon cancels 241 phrase-boundary candidates; v9's cancels 3. Last row: a *fresh* 2×-larger stretch set (seed changed, 50 stretches, 184 boundaries) drawn after all development ended — latency and false fires hold; recall dips 4.5 pp (§10). †WER mean dominated by long-monologue repetition loops when no force-flush bounds the segment (v9 fresh-set WER *median* 0.30, matching the development set; the §7 force-flush bounds this tail).

Policy	Recall ↑	P50 ↓	P95 ↓	False/min ↓	WER_{mean} ↓
v5 + gate + confirm1	0.927	0.68 s	1.27 s	3.2	0.36
v8 + gate + confirm1	0.938	0.77 s	1.06 s	0.0	4.96†
v9 + gate	**0.969**	**0.39 s**	**0.70 s**	**0.3**	**1.43**
v9 + gate + confirm1	0.969	0.89 s	1.20 s	0.0	1.43
v9 + gate (fresh 50-stretch set)	0.924	0.42 s	0.70 s	0.2	4.63†

differing only in an observable feature — precisely the feature the model must learn to key on. (2) Holding through a pause is trained only where the prefix itself says the utterance is incomplete (trailing filler/connective, mid-word cut); pauses after complete speech fire even if the speaker later resumes — quick resumes become two segments, the benchmark's measured cost, not an error. Eight schemas cover fire, hold, truncation, pause discrimination, pure silence, and silence-initial segments (Appendix A).

The fingerprint, removed. Figure 1 (right): the composite climbs monotonically 0.900→0.963 and plateaus; fire and hold classes sit at ceiling simultaneously for the entire run. Architecture, recipe, scale, and sources held fixed — as close to a controlled experiment on the label spec as a training run gets.

Dominance. Table 2 and Figure 7. Against its predecessors, v9+gate is better on every axis at once. Against the deployed cascade, the comparison's *structure* is the point: a timeout must be long enough to survive within-turn pauses, so it can only trade latency against false fires along one curve; v9 escapes the curve by reading completeness. And the crutches retire: the phrase-vs-turn discrimination moved from inference policy into weights — which is what

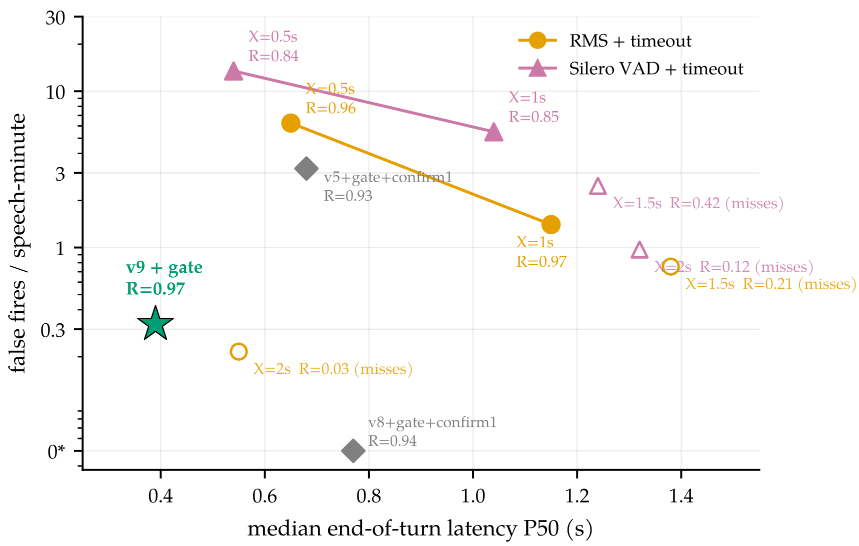

Figure 7. Semantic endpointing escapes the timeout curve. Median latency vs false fires (log scale; 0* = zero plotted at 0.05) on identical stretches, detector, and scoring. A silence timeout X can only slide along its curve: matching v9's recall takes X=1.0 s — 3× the latency, ∼5× the false fires; the fastest usable setting (X=0.5 s) produces 21× the false fires; $X \geq 1.5$ s leaves the recall tolerance window entirely (open markers) — the assistant that waits too long. The Silero cascade [Silero Team, 2021] is dominated by plain RMS on this close-talk channel. v9 sits strictly inside the whole family because it reads *completeness*: it fires 0.39 s after a finished thought and holds through a 1.4 s mid-sentence pause. Honest caveat: v9 segments quick resumes at the same rate as an aggressive timeout (resume-after-fire 1.0, by design — the correct choice is unknowable at decision time); deployments preferring merged segments pay +0.5 s via confirm1.

supervision was supposed to do all along.

6 The rest of the model still works

An endpoint fine-tune that damaged the base recognizer would be a bad trade. It does not — and the unified model's side capability comes with numbers its base model never published.

Offline ASR regression — measured, attributed, and honest. Table 3: the endpoint fine-tune costs +1.1 pp WER on LibriSpeech test-clean and +2.7 pp on test-other. Our first hypothesis was mechanical: the model fires its marker on 4.7–5.9% of offline clips (many LibriSpeech clips carry the ≥0.3 s silence tail where the v9 rule *says* fire), and a fire ends the decode, truncating the transcript. We tested it by banning the two marker tokens at decode time — and the hypothesis *fails*: fires drop to zero, WER does not move (3.94%/7.91%). The regression is genuine distributional drift from fine-tuning on ∼12k narrow-schema examples with no plain-ASR replay data, and the standard mitigation (mixing transcription-only data into the fine-tune) is unexercised in this release. The deployment-relevant *streaming* WER is unaffected (Table 2, median 0.28–0.30 across sets), so the cost is confined to using the endpoint checkpoint as an offline transcriber — a job the base checkpoint still does.

Context biasing, measured where it matters. Figure 8(a): on Earnings-22 [Del Rio et al., 2022] — spontaneous earnings-call speech dense in tickers, executives, products; entities LLM-

Table 3. Offline WER regression (LibriSpeech, full official splits, single-shot decode on vLLM, jiwer/whisper normalization). "Fires" = fraction of clips where the model emitted the endpoint marker (offline clips end at or near end-of-speech; base never fires because its marker rows are untrained).

	WER↓		Fires	
Model	clean	other	clean	other
Qwen3-ASR-0.6B (base)	2.79%	5.14%	0%	0%
+ endpoint LoRA (merged v9)	3.93%	7.87%	4.7%	5.9%
+ markers banned at decode	3.94%	7.91%	0%	0%

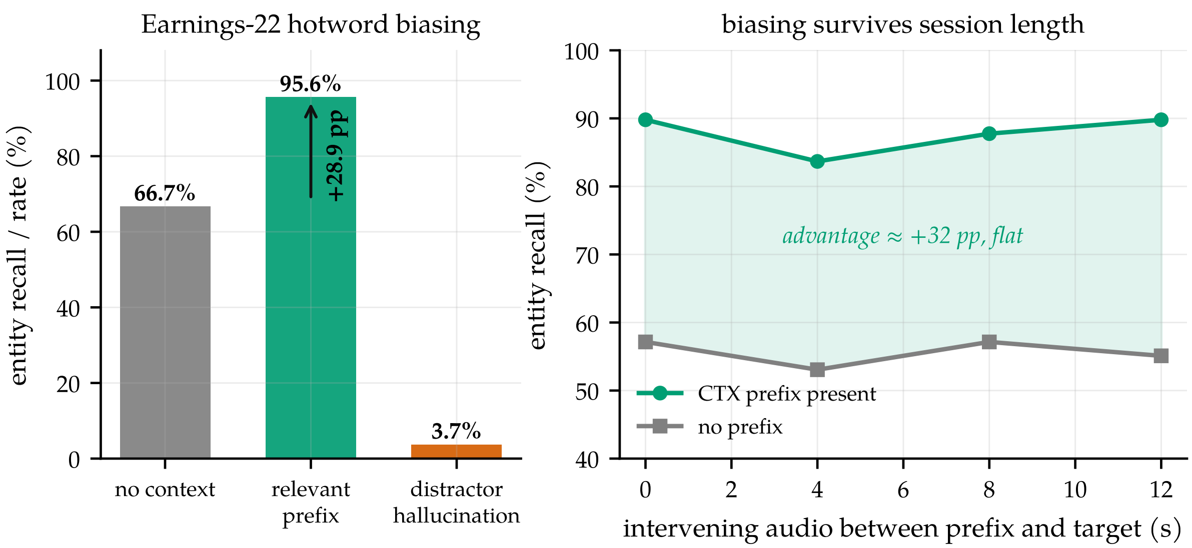

Figure 8. Context biasing: the capability with no public numbers. (a) Earnings-22 entity recall without context, with the relevant hotword prefix, and the hallucination rate under a distractor prefix (150 utterances; LLM-extracted proper-noun entities). (b) Recall vs seconds of unrelated speech interposed between prefix and target: the ≈+30 pp advantage does not decay with session age.

extracted and filtered to genuine proper nouns — a relevant hotword prefix lifts entity recall from 66.7% to 95.6% (+28.9 pp), while nudging WER down (15.3→15.0%). A *distractor* prefix (another utterance's entities) induces 3.7% hallucination — above our 3% target, reported as-is; distractor-ratio training is the known, unexercised lever. Qwen3-ASR claims context biasing but reports no numbers; to our knowledge these are the first on this model family.

The advantage survives session length. Figure 8(b): with 0–12 s of unrelated speech interposed between the prefix and the target words, the biasing advantage stays flat at ≈+30 pp. Context biasing is a durable session-level capability, not a first-utterance parlor trick.

7 Deployment reality check

Streaming papers routinely stop at a simulator. Three findings from serving this model on stock vLLM [Kwon et al., 2023] (one *shared* GPU throughout):

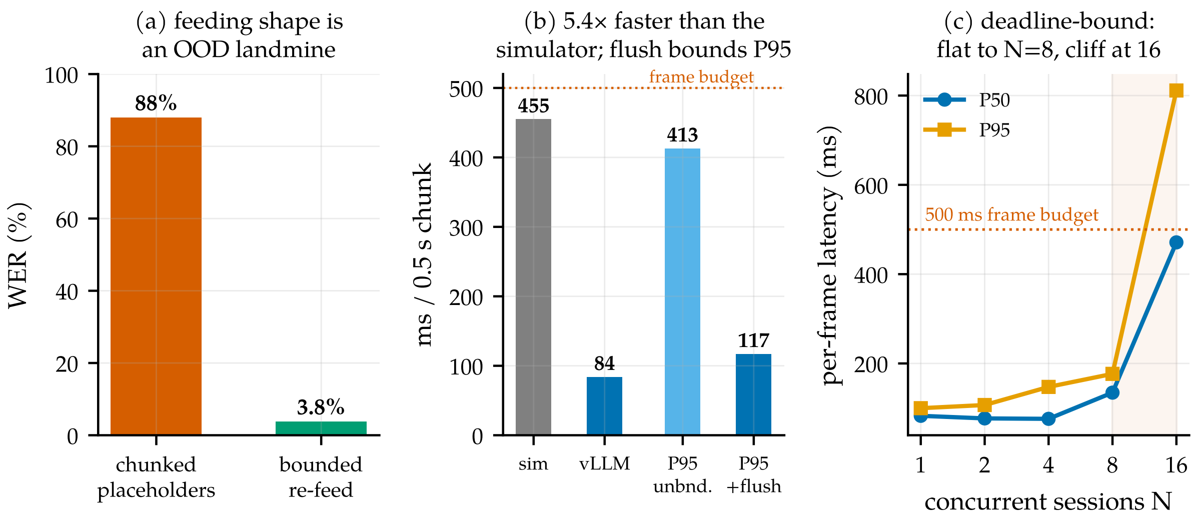

Figure 9. Serving on real infrastructure. (a) Feeding shape: per-chunk audio placeholders are out-of-distribution for a model trained on single segments (88% WER); bounded re-feed of the last window costs a re-encode but is correct (3.8%). (b) Per-chunk compute: the vLLM path is 5.4× faster than the transformers simulator used for all pre-serving latency numbers; the max-segment force-flush also bounds the P95 tail. (c) Concurrent sessions per engine on a 0.15-GPU slice: flat to N=8, cliff at 16 — the deadline binds, not memory.

An OOD landmine. Figure 9(a). The natural flat-cost serving shape — append each 0.5 s chunk as a new audio placeholder in a growing resident request — destroys the model: 88% WER vs 3.8% for re-feeding the last window as a single segment. The base model was trained on single audio segments; per-chunk placeholders are wildly out of distribution. Bounded re-feed is the only correct primitive for this family.

The simulator was the bottleneck. Figure 9(b). The replay protocol on vLLM reproduces the endpoint behavior at 84 ms median per chunk vs the transformers simulator's 455 ms (5.4×) — ∼6× real time on a 0.15-utilization slice. Latency claims made on $O(T^2)$ re-decode simulators overstate the cost of this approach by that factor.

Bound every resident state. Unbounded committed prefixes eventually loop on long monologues; a 20 s force-flush fixes the WER tail (2.54→1.29) *and* the compute tail (P95 413→117 ms) at no endpoint cost (Figure 9(b)). Under concurrency (Figure 9(c)) per-frame p95 stays flat through N=8 sessions per 0.15-GPU slice and cliffs at 16 — deadline-bound, not memory-bound, with the energy gate batching only the sessions actually due each frame. (A contention-caveated lower bound; the shape, not the constant, is the result.) The in-engine efficiency variant — windowed KV with the `<CTX>` prefix pinned as an attention sink [Xiao et al., 2024] — is validated separately in a companion systems paper [Li, 2026]; capability-wise, bounded re-feed already keeps biasing flat (Figure 8(b)).

8 Clairvoyant labels: a checklist

The failure mode generalizes to any model that must act mid-stream on offline-labeled data. Four questions for a streaming supervision pipeline:

Is your supervision clairvoyant?

1. **Prefix test.** For each label at decision time t: is it computable from input up to t? *(Our pause labels were keyed on the next speaker.)*
2. **Twin test.** Can two causally identical prefixes receive different labels anywhere in the data? *(Same complete-utterance prefix: fire in one pool, hold in the other.)* Minimal pairs differing in an *observable* are the fix, not the bug.
3. **Phantom-condition test.** Does the eval demand behavior on conditions deployment never produces? *(Clips cut at end-of-speech; deployment always delivers the next silence.)*
4. **Oscillation test.** Does training bounce between mode attractors that each satisfy part of the data? That is not instability to be scheduled away; it is the loss surface reporting a contradiction.

Candidates we believe are affected: turn-taking policies for full-duplex agents trained from offline dialogue transcripts (the labels encode who *did* speak next, not what was knowable — cf. the projection-style objectives of Ekstedt and Skantze, 2022, which are explicitly predictive and thus honest about the uncertainty); barge-in and endpointing in commercial stacks; simultaneous translation trained with reference-aligned read/write decisions [Ma et al., 2019]; streaming TTS interruption; tool-call timing for agents cloned from completed trajectories — a streaming sibling of causal confusion in imitation learning [de Haan et al., 2019]. In each, an apparent aggressiveness-vs-caution frontier may partly be an artifact of asking the model to know the future. The diagnosis also joins a broader lineage of measurement-induced phenomena: as sharp metric choices can manufacture "emergence" [Schaeffer et al., 2023], non-causal label choices can manufacture *trade-offs*.

9 Related work

Endpointing and end-of-turn detection. Classical endpointing thresholds a VAD [Silero Team, 2021] with a silence timeout; learned refinements predict end-of-query directly from acoustics [Shannon et al., 2017], jointly train an endpointer with the recognizer [Chang et al., 2019, Li et al., 2020, Bijwadia et al., 2023], or distill tiny acoustic end-of-turn models [Helwani et al., 2026, Pipecat team, 2025]. Recent systems fuse acoustic and linguistic cues for the turn decision [Wang et al., 2026, Yang et al., 2026] or fold VAD, turn detection, and ASR into one audio-LLM front-end [Li et al., 2026]. Commercially, Deepgram Flux integrates end-of-turn events into the recognizer's forward pass [Deepgram, 2026] and OpenAI exposes semantic VAD [OpenAI, 2025]; neither publishes training details. Our contribution to this line is orthogonal to architecture: a supervision principle, with evidence that violating it manufactures the recall-precision trade-offs this literature often reports as intrinsic.

Turn-taking in dialogue. Turn-taking prediction has a long tradition in conversational AI [Skantze, 2021], from text-based end-of-turn prediction [Ekstedt and Skantze, 2020] to voice-activity projection trained on future windows [Ekstedt and Skantze, 2022]. Projection objectives predict distributions over the future and are therefore causally honest; our diagnosis concerns

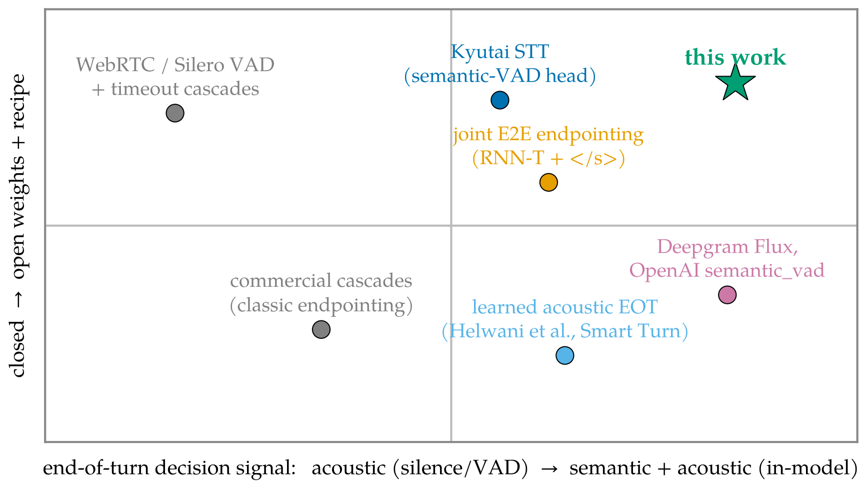

Figure 10. Where this work sits. End-of-turn systems by decision signal (acoustic silence vs. in-model semantics) and openness. The semantic quadrant's existing members are closed products; the open systems are acoustic cascades, weights-only releases, or joint E2E endpointers without a published turn-training recipe. This work occupies the empty corner: semantic end-of-turn, open weights *and* recipe.

the sharper failure where future-dependent quantities are presented as *deterministic labels*.

Streaming and unified ASR. Streaming recognition spans RNN-T [Graves, 2012], large weakly supervised models [Radford et al., 2023], decoder-only streaming with latency losses [Wan et al., 2026], and unified streaming/offline speech-LLMs [Shi et al., 2026]. Kyutai's delayed-streams models add a semantic-VAD head to streaming STT [Kyutai Labs, 2025, Défossez et al., 2024]. We inherit this substrate and add the turn event plus the training recipe.

Context biasing. Contextual ASR ranges from attention-based deep context [Pundak et al., 2018] to retrieval-scale biasing [Gong et al., 2025], RL-tuned hotword integration [Kong et al., 2025], and cue-based prompting for speech LLMs [Novitasari et al., 2026]. We use plain prompt-slot biasing and contribute measurements: uplift on entity-dense audio and durability across session age.

Serving. vLLM's paged KV [Kwon et al., 2023] carries our serving path; attention sinks [Xiao et al., 2024] and bounded-state streaming serving [Li, 2026] provide the in-engine efficiency variant of our pinned context prefix.

10 Limitations

Single-speaker, single-channel English; the mixed-channel meeting shape is future work. Semantic completeness labels come from a heuristic transcript classifier (fillers, connectives, short-utterance rules); a learned completeness judge is the obvious next lever on the residual recall gap. Distractor hallucination (3.7%) exceeds our 3% target. Quick resumes are always segmented (resume rate 1.0); merging them costs +0.5 s via the confirm knob or requires post-hoc fusion. Concurrency numbers are contention-caveated lower bounds from a shared GPU. The 25-stretch benchmark (96 turn-final boundaries) was reused across development, so we drew a fresh 2×-larger set (seed changed, 50 stretches, 184 boundaries) after all development ended: latency, false fires, and WER median reproduce, while recall drops 0.969→0.924 (Table 2) — consistent with mild adaptive overfitting to the development set plus a harder draw; we report both and treat the fresh set as the honest operating estimate. The endpoint fine-tune also costs offline WER (§6): mixing plain ASR data into the fine-tune is the standard, unexercised mitigation.

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A The v9 label specification

Eight schemas, ∼12k examples, LibriSpeech and AMI roughly 50/50; pair examples use both same- and different-speaker AMI pairs (the fire decision keys on completeness + silence, never on speaker identity, which single-channel deployment cannot observe). M = <EAGER_END_SPEECH><END_SPEECH>.

Table 4. v9 training schemas. Schemas 1–2 are the minimal-pair device (Figure 6); 7–8 fix silence incompetence (Figure 4).

#	Schema	Construction	Target
1	`single_sil`	utterance + 0.3–1.2 s silence	`text` M
2	`complete_nosil`	*same utterances as #1*, tail ≤0.1 s	`text`
3	`truncated`	cut mid-speech at 30–80%	partial `text`
4	`pair_fire`	complete A + gap 0.3–2.5 s + B (+ tail)	`A` M `B` M / `A` M `B`
5	`pair_hold`	*incomplete* A + gap + B + tail	`A B` M
6	`long_sil`	utterance + 2–4 s silence	`text` M
7	`silence_only`	pure silence 0.5–4 s	(empty)
8	`lead_sil`	silence 0.5–2 s + utterance + tail	`text` M

Completeness classifier for pair examples: utterance A is *incomplete* iff its transcript ends in a filler/connective ("and, but, so, or, uh, um, the, a, to, of, in, with, i mean, you know, . . . "), ends mid-word (AMI partial-word annotation), or has fewer than 3 words outside a closed acknowledgment list. Borderline cases land in the `resume_after_fire` cost knob — they never create fire/hold contradictions on identical observables. Abandonment (incomplete speech, speaker never resumes) is an inference-policy timeout, not a label.

B Benchmark specification and the spec-v1 bug

Full protocol: 0.5 s chunks; committed-prefix incremental decoding (emitted text committed, last 5 tokens rolled back) matching the base model's official streaming semantics; fires timestamped at the chunk boundary where <END_SPEECH> first appears; recall window $[-0.25, +1.5]$ s clipped at the next utterance's onset. The first version of the boundary classifier promoted a boundary to turn-final when another meeting speaker interleaved before the target speaker resumed. On a single-channel stream that speech is silence — demanding a fire there is exactly the clairvoyance the benchmark exists to remove. The corrected (spec-v2) classification moved v5 recall 0.917→0.906 and v8 0.935→0.938; no conclusion changed, but the episode is a working example of §8 applied to one's own evaluation.

C Additional results

Gated arms and the confirm-horizon sweep. Table 5. The confirm horizon behaves as a clean latency-vs-false-fire dial: h=1 strictly improves v5 (recall *rises* because deferring the fire lets the segment gather the boundary inside tolerance) and zeroes v8's and v9's false fires; h=2 over-suppresses (v5 recall 0.812, latency past budget). v9 is the only checkpoint whose h=0 row is already deployable. Ungated arms are omitted as uninterpretable (Figure 4): with ∼1.9 spurious fires per second on silence, v3/v5/v8 score 0.95–0.99 "recall" by chance; their gated rows below are the meaningful ones.

Table 5. All gated arms on the 25-stretch benchmark. Resume = `resume_after_fire` rate over continuation boundaries (a cost knob, not an error rate).

Arm	Recall	P50	P95	False/min	Resume	$\mathrm{WER}_{\mathrm{mean}}$
v3 + gate	0.896	0.05 s	0.25 s	88.8	0.89	0.85
v5 + gate	0.906	0.16 s	0.89 s	27.3	0.96	0.36
v5 + gate + confirm1	0.927	0.68 s	1.27 s	3.2	0.85	0.36
v5 + gate + confirm2	0.812	1.15 s	1.42 s	1.1	0.56	0.36
v8 + gate	0.938	0.26 s	0.56 s	3.6	0.93	4.96
v8 + gate + confirm1	0.938	0.77 s	1.06 s	0.0	0.85	4.96
v9 + gate	0.969	0.39 s	0.70 s	0.3	1.00	1.43
v9 + gate + confirm1	0.969	0.89 s	1.20 s	0.0	0.93	1.43

The full timeout family. Table 6, both detectors on the original (seed 0, 25 stretches) and fresh (seed 1, 50 stretches) sets. The family's shape is stable across sets: recall collapses between X=1.0 and 1.5 s (the fire lands outside the $+1.5$ s tolerance), and no setting reaches v9's (recall, latency, false-fire) point. Silero ran per-chunk with state resets — a slight handicap noted in §5; RMS shares the LM arms' exact detector and is the apples-to-apples policy comparison.

Table 6. Silence-timeout cascade, all settings and both stretch sets.

Detector	Set	X (s)	Recall	P50	P95	False/min	Resume
RMS	seed 0	0.5	0.958	0.65 s	0.94 s	6.3	1.00
RMS	seed 0	1.0	0.969	1.15 s	1.44 s	1.4	0.81
RMS	seed 0	1.5	0.208	1.38 s	1.49 s	0.8	0.07
RMS	seed 0	2.0	0.031	0.55 s	0.55 s	0.2	0.00
RMS	seed 1	0.5	0.978	0.70 s	0.95 s	4.6	0.99
RMS	seed 1	1.0	0.989	1.20 s	1.45 s	0.6	0.73
RMS	seed 1	1.5	0.185	1.44 s	1.50 s	0.4	0.07
RMS	seed 1	2.0	0.016	1.14 s	1.14 s	0.2	0.00
Silero	seed 0	0.5	0.844	0.54 s	0.88 s	13.5	0.96
Silero	seed 0	1.0	0.854	1.04 s	1.38 s	5.5	0.85
Silero	seed 0	1.5	0.417	1.24 s	1.49 s	2.5	0.41
Silero	seed 0	2.0	0.125	1.32 s	1.42 s	1.0	0.00
Silero	seed 1	0.5	0.837	0.51 s	0.83 s	12.3	0.88
Silero	seed 1	1.0	0.864	1.00 s	1.33 s	4.4	0.73
Silero	seed 1	1.5	0.451	1.29 s	1.48 s	2.0	0.27
Silero	seed 1	2.0	0.130	1.34 s	1.44 s	1.3	0.07

Fresh-set confirmation. v9+gate on the seed-1 set: recall 0.924, P50 0.42 s, P95 0.70 s, 0.21 false fires/min, resume 0.91, WER median 0.30 / mean 4.63 (three long-monologue stretches enter the no-flush repetition loop; the §7 force-flush is the documented fix). Read against the seed-1 timeout rows above: the cascade can reach higher recall on this set (0.978–0.989 at $X \in \{0.5, 1.0\}$ s) but only at 1.7–2.9× v9's latency and 3–22× its false fires — and no timeout setting reaches v9's $(0.42\,\mathrm{s}, 0.21/\mathrm{min})$ corner at *any* recall. The 4.5 pp recall drop from the development set is an honest generalization gap discussed in §10.

D Reproducibility

Hardware: one RTX Pro 6000 (Blackwell), shared with an unrelated co-tenant workload throughout (all GPU jobs wrapped in retry/resume; the v9 training run survived four OOM kills via checkpoint resume with $\leq$750 steps lost each). Training: LoRA r=16, α=32, batch 8, best checkpoint at step 7.5k selected by the holdout composite. Evaluation and serving: vLLM 0.19, `skip_special_tokens=False` (the markers are reserved special tokens and are otherwise silently stripped — a gotcha worth one sentence). Code, the label-spec builder, the replay benchmark, and the merged checkpoint are released at `https://github.com/19PINE-AI/turn-aware-asr`.